
Intent Stabilization for Ambiguous User Requests

Pranav S. Dulepet*
Johns Hopkins University
pdulepe1@jh.edu

Asmitha Sathya*
Johns Hopkins University
asathya7@jh.edu

May 3, 2026

Abstract

A common failure mode in assistant-like AI usecases is ambiguous requests from the user. Assistants usually do not understand the request and misinterpret it or even answer incorrectly. We study *intent stabilization* by sampling structured intent from user requests. We then repair the model understanding path when it is unstable and clarify when there is still uncertainty left. The method is inference time and is not intrusive, so it works with any model. On InfoQuest and ClarifyBench test splits with Qwen3-30B and a Llama-3.3-70b judge model, we find that all-stage resampling matches generic clarification on InfoQuest task-success rate (0.240) and slightly improves upon targeted clarification (0.235), but falls behind the highest scoring ClarifyBench clarification baseline (0.500 vs. 0.544). The K-sensitivity runs show that more samples are not monotonically better: $K = 7$ gives the best dev task-success rate (0.240) but costs 26.95 task-model calls per example, while $K = 1$ is close (0.230) at 6.07 calls. We find that structured resampling is useful for diagnosing unstable intent, but simple clarification is still a strong quality-cost baseline.

1 Introduction

The quality of AI assistants should be measured by how well it understands the user’s intent. AI models are now great at generating text in a fluent, human-like way, but the gap remaining is whether they understand ambiguous requests as well as humans do.

We study how before committing to an answer or a clarification, whether sampling multiple structured interpretations of the request helps AI model intent recognition. Our proposed wrapper breaks a request down into three stages: ambiguity assessment, candidate intent modeling, and response strategy. It samples each stage, aggregates any disagreement, does an all-stage resampling when the state is unstable, and asks a clarification if uncertainty remains high. We call this procedure *intent stabilization*.

Our research question is: does sampling-based stabilization improve ambiguity handling enough to justify its extra model calls? On InfoQuest, the all-stage wrapper matches generic clarification and slightly beats targeted clarification. On ClarifyBench, it remains safe but underperforms the best clarification baseline. The ablations and K-sensitivity runs show that resampling can help, but cost and calibration are still major problems.

The primary contributions are:

- A model-agnostic intent-stabilization wrapper that uses structured multi-sampling rather than token-level internals.

*Equal contribution.

- Frozen ambiguous-request evaluation splits for InfoQuest and ClarifyBench with automatic validation and judge-based scoring.
- A comparison against direct answering, hedged answering, generic clarification, and targeted clarification baselines.
- An ablation analysis showing which stabilization choices matter and a cost analysis showing why all-stage resampling is not yet a drop-in replacement for simple clarification.

2 Related Work

Clarification and ambiguous question answering. Qulac formulates clarification in open-domain information-seeking conversations and shows that a single useful clarification can substantially improve retrieval quality [1]. ClariQ extends this line into a shared benchmark for ranking clarifying questions in ambiguous dialogue settings [2]. AmbigQA studies open-domain questions with multiple plausible interpretations and evaluates systems by requiring disambiguated question-answer pairs [13].

Pragmatic intent and user modeling. The method is also related to pragmatic accounts of communication, where listeners infer intended meaning from context. Rational Speech Act models formalize pragmatic language interpretation as probabilistic inference over speaker meaning [7], and recent surveys argue that grounded language systems need common ground, task goals, and context to figure out ambiguity [5]. User-modeling work makes the same point: the Lumiere project used Bayesian user models to infer software users’ goals and needs from actions and queries [9], while human-AI collaboration work shows that agents perform better when they explicitly model human behavior or goals [3, 12, 17].

Sampling-based reasoning and uncertainty. Chain-of-thought prompting shows that intermediate reasoning traces can improve large-model reasoning [15]; self-consistency then improves robustness by sampling multiple reasoning paths and selecting the answer supported by the most paths [14]. Recent uncertainty work makes a similar move from individual strings to meaning-level agreement. Semantic entropy clusters sampled generations by semantic equivalence to detect confabulations in free-form generation [4].

Calibration and selective prediction. Calibration asks whether a model’s stated confidence corresponds to empirical correctness [8]. Selective classification studies systems that may abstain when confidence is too low, trading coverage for lower risk [6]. For language models, verbalized confidence and self-evaluation have been studied as ways to expose uncertainty without direct access to model internals [11, 10].

LLM-based evaluation. Open-ended assistant outputs are difficult to score with heuristic and exact/fuzzy-match metrics. LLM-as-judge evaluation has been used as a scalable approximation to human preference judgments, while also introducing bias and reliability concerns [16].

3 Datasets

We evaluate on two frozen ambiguous-request datasets. InfoQuest is a 1,000-example collection of underspecified information-seeking requests split into 600 train, 200 development, and 200 test examples, grouped by seed identifier so related variants do not cross splits. The final results use the 200-example test split.

ClarifyBench v1 is a 120-example in-repository benchmark designed for multi-turn clarification behavior. It covers four ambiguity types—lexical, missing context, referential, and underspecified requests—across five domains: finance, health, shopping, software, and travel. We use the 90-example test split as a held-out generalization check. Each example has a simulated user reply and a gold answer so that clarification policies can be evaluated after the follow-up turn.

Both datasets are validated before report runs for schema completeness, duplicate identifiers, split membership, seed grouping, and metadata coverage.

4 Methods

All methods are inference-time policies over the same task model. The first four are baselines; the fifth is the proposed intent-stabilization wrapper.

- **Direct answer:** answer the user request immediately without ambiguity handling.
- **Generic hedge:** answer directly while using hedging language.
- **Generic clarify:** ask a broad clarification question before giving a final answer after the simulated user reply.
- **Targeted clarify:** run a structured ambiguity detector and intent modeler, then ask a more targeted clarification question before the final answer.
- **All-stage intent stabilization:** sample multiple ambiguity assessments, intent hypotheses, and response strategies; aggregate agreement; resample the full stage stack if the state is unstable; and ask a clarification if uncertainty remains.

All-stage intent stabilization. Given a request x , the wrapper draws K structured samples for the ambiguity, intent-modeling, and response-strategy stages. Each sample is parsed into a schema. The aggregator measures whether samples agree that the request is ambiguous, converge on the same plausible intents, and recommend the same action. These agreement features become the reported intent-confidence score.

Formally, the wrapper treats the task model as an implicit sampler over latent interpretations. For sample j , it draws an ambiguity assessment $a_j \sim q_\theta(a \mid x)$, an intent hypothesis $i_j \sim q_\theta(i \mid x, a_j)$, and a response strategy $s_j \sim q_\theta(s \mid x, a_j, i_j)$. The empirical bundle $\{(a_j, i_j, s_j)\}_{j=1}^K$ approximates uncertainty over user intent without requiring direct access to $P_\theta(i \mid x)$. Stability is then measured by agreement across sampled structured outputs rather than by the model’s verbal confidence alone.

If agreement is below threshold, the wrapper spends a small repair budget to resample the full ambiguity-intent-strategy bundle. If the repaired state is still unstable, it asks a clarification question rather than answering directly. In these benchmarks, clarification is the safe fallback because every evaluation example is intentionally ambiguous.

Why all-stage repair? The ablation suite compares single-sample behavior, no clarification fallback, one repair round, resampling all stages, and related variants. The strongest variant resamples the full state together. This result is consistent with the structure of the task: ambiguity judgments, candidate intents, and strategy decisions are coupled. Repairing them independently can leave the controller with internally inconsistent evidence.

5 Evaluation

The model in the our runs is Qwen/Qwen3-30B-A3B-Instruct-2507. We use a separate judge model, meta-llama/Llama-3.3-70B-Instruct, so that the task model is not judging its own outputs. We use Tinker with deterministic final-answer settings where supported and stochastic sampling for the intent-stabilization stages.

Metrics. We report task-success rate (TSR), appropriate-action rate (AAR), final answer score, clarification quality, task-model calls, latency, and resampling rate. TSR is the main end-task metric. After the policy acts and, when applicable, receives the simulated user reply, the final answer must be similar to the gold target. AAR measures whether the policy chose a safe action under ambiguity; because the test sets are ambiguous on purpose, direct answering scores a low AAR even when the answer seems fluent on the surface-level. Clarification quality measures whether the clarification question helps resolve the missing intent rather than asking a high-level, simple follow-up.

Interpretation. These metrics separate safety from usefulness. A method can score a perfect AAR by always clarifying and still fail to improve TSR if the clarification is generic or the final answer is still weak. Direct answering can also sometimes be the correct answer but can be penalized for unsafe behavior under ambiguity. This is why we frame the question as whether intent stabilization improves final success beyond simple clarification and whether any gain justifies its additional calls.

Table 1: Main results on frozen test splits with Qwen3-30B as the task model. TSR is task-success rate; AAR is appropriate-action rate.

Dataset	Method	TSR	AAR	Answer	Clarify
InfoQuest	Direct answer	0.065	0.000	0.218	0.000
InfoQuest	Generic hedge	0.025	0.000	0.208	0.000
InfoQuest	Generic clarify	0.240	1.000	0.499	0.658
InfoQuest	Targeted clarify	0.235	1.000	0.482	0.329
InfoQuest	All-stage stabilization	0.240	1.000	0.492	0.655
ClarifyBench	Direct answer	0.144	0.000	0.158	0.000
ClarifyBench	Generic hedge	0.022	0.000	0.031	0.000
ClarifyBench	Generic clarify	0.522	1.000	0.627	0.566
ClarifyBench	Targeted clarify	0.544	1.000	0.627	0.230
ClarifyBench	All-stage stabilization	0.500	1.000	0.609	0.578

Table 2: All-stage stabilization ablations on InfoQuest test with Qwen3-30B.

Ablation	TSR	AAR	Clarify	Calls	Resample
Clarify immediately	0.245	1.000	0.376	12.00	0.000
Generic question fallback	0.245	1.000	0.659	14.09	0.955
No calibration	0.245	1.000	0.445	14.12	0.950
No clarification fallback	0.215	1.000	0.464	14.14	0.940
No selective resample	0.255	1.000	0.578	14.79	0.930
One round only	0.245	1.000	0.446	14.10	0.930
Resample all stages	0.265	1.000	0.472	14.82	0.940
Single sample only	0.225	1.000	0.560	6.00	0.000

6 Results

Table 1 gives the main comparison on the Qwen3-30B task model. Direct answering and hedged answering perform poorly on both datasets because all evaluation examples require ambiguity handling. Generic and targeted clarification are strong baselines: they always choose a safe action and often recover a useful final answer after the simulated follow-up.

All-stage intent stabilization is competitive on InfoQuest, matching generic clarification at 0.240 TSR and slightly exceeding targeted clarification at 0.235. On ClarifyBench, however, it goes to 0.500 TSR, below generic clarification at 0.522 and targeted clarification at 0.544. This is the main empirical conclusion: resampling stabilizes the controller’s action choice, but the current policy does not reliably turn extra intent samples into better final answers across datasets.

Ablations. Table 2 isolates the resampling design. The strongest ablation is the all-stage repair variant, which reaches 0.265 TSR on InfoQuest while preserving 1.000 AAR. Single-sample behavior is weaker at 0.225, and removing clarification fallback is worse at 0.215. Calibration changes alone do not explain the result. The evidence supports all-stage repair as the best variant of the wrapper, even though the confirmed two-dataset result remains cost-heavy and mixed.

Cost, confidence, and K . The confirmed all-stage run uses 14.88 task-model calls per InfoQuest example and 13.80 per ClarifyBench example, compared with roughly two calls for simple clarification. Table 3 shows that increasing K is not a free improvement: $K = 7$ has the best dev TSR (0.240) but costs 26.95 calls, while $K = 1$ is close (0.230) at 6.07 calls. Confidence is also not calibrated as final-answer correctness. For example, $K = 1$ reports mean confidence 0.957 but only 0.230 TSR, whereas $K = 7$ reports lower mean confidence 0.639 with slightly higher TSR. We therefore use confidence as a conservative control signal, not as a probability of being correct.

Table 3: Matched K-sensitivity for all-stage stabilization.

K	Repair	Rounds	TSR	AAR	Conf.	Calls
1	1	1	0.230	0.995	0.957	6.07
3	1	1	0.225	1.000	0.681	14.85
5	1	1	0.205	1.000	0.649	21.00
7	1	1	0.240	1.000	0.639	26.95

7 Limitations

The primary limitation is that all reported scores are LLM-as-a-judge-based. A different judge model reduces direct self-evaluation bias, but it does not replace a human audit since LLMs are not equivalent to humans. The benchmark design also makes clarification the safe action for every test example, so appropriate-action rate is easier for always-clarify methods than it would be in a mixed benchmark containing many unambiguous requests.

The method is also expensive. All-stage stabilization makes around seven times more task-model calls than simple clarification on InfoQuest. This cost is only justified if the additional samples produce better final answers, which the current results do not consistently represent. We also use agreement among structured text samples rather than token log probabilities, hidden states, or retrieval-grounded evidence for the sake of reproducibility, simplicity, and model-agnosticism. This may cause us to miss uncertainty signals that are only seen inside open-weight models.

The current policy also tends to clarify after resampling rather than confidently resolving ambiguity without a user turn. The next step should be a better policy for converting stable intent hypotheses into targeted clarifying questions or high-confidence narrowed answers.

The confidence estimator should also be interpreted cautiously. The current score is derived from agreement among structured samples and is useful for deciding when to avoid direct answering, but the K-sensitivity results show that it is not calibrated as a probability of final correctness. The sensitivity study varies K at fixed repair samples, repair rounds, threshold, and temperature; a full grid over all controller parameters remains future work.

8 Reproducibility

The repository includes frozen data splits, deterministic split manifests, resumable inference, and per-run metrics files. Reported task-model runs use Qwen3-30B; judge runs use Llama-3.3-70B.

The most important reproducibility caveat is non-determinism. Even at low final-answer temperature, we can get small variations through sampling, retries, and model-infrastructure changes.

9 Conclusion

All-stage resampling is a useful diagnostic for unstable intent even if it is not a clear replacement for simple clarification. It matches the best InfoQuest clarification baseline, underperforms on Clarify-Bench, and costs substantially more calls. Future work should focus more on using sampled intent hypotheses to ask better targeted clarifications or produce cheaper, calibrated control decisions.

References

- [1] Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 475–484, 2019. doi: 10.1145/3331184.3331265.
- [2] Mohammad Aliannejadi, Julia Kiseleva, Aleksandr Chuklin, Jeff Dalton, and Mikhail Burtsev. Convai3: Generating clarifying questions for open-domain dialogue systems (clariq). *arXiv preprint arXiv:2009.11352*, 2020.

- [3] Micah Carroll, Rohin Shah, Mark K. Ho, Thomas L. Griffiths, Sanjit A. Seshia, Pieter Abbeel, and Anca Dragan. On the utility of learning about humans for human-ai coordination. In *Advances in Neural Information Processing Systems*, 2019.
- [4] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024. doi: 10.1038/s41586-024-07421-0.
- [5] Daniel Fried, Nicholas Tomlin, Jennifer Hu, Roma Patel, and Aida Nematzadeh. Pragmatics in language grounding: Phenomena, tasks, and modeling approaches. *arXiv preprint arXiv:2211.08371*, 2022.
- [6] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.
- [7] Noah D. Goodman and Michael C. Frank. Pragmatic language interpretation as probabilistic inference. *Trends in Cognitive Sciences*, 20(11):818–829, 2016. doi: 10.1016/j.tics.2016.08.005.
- [8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330, 2017.
- [9] Eric Horvitz, Jack Breese, David Heckerman, David Hovel, and Koos Rommelse. The lumiere project: Bayesian user modeling for inferring the goals and needs of software users. In *Proceedings of the Fourteenth Conference on Uncertainty in Artificial Intelligence*, pages 256–265, 1998.
- [10] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- [11] Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022.
- [12] Chang Liu, Jessica B. Hamrick, Jaime F. Fisac, Anca D. Dragan, J. Karl Hedrick, S. Shankar Sastry, and Thomas L. Griffiths. Goal inference improves objective and perceived performance in human-robot collaboration. In *Proceedings of the 15th International Conference on Autonomous Agents and Multiagent Systems*, pages 940–948, 2016.
- [13] Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. Ambigqa: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 5783–5797, 2020.
- [14] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- [15] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [16] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023.
- [17] Tan Zhi-Xuan, Lance Ying, Vikash Mansinghka, and Joshua B. Tenenbaum. Pragmatic instruction following and goal assistance via cooperative language-guided inverse planning. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, 2024.